Temporal & Multimodal Visual Intelligence

Technical research report | Six-class HMDB51 action recognition | Experimental results verified against saved metrics

1 Research question and central finding

Can a model use information from a sequence of visual observations that is absent from one frame? I compare one-frame recognition, unordered eight-frame aggregation, and ordered recurrent modeling under a common frozen DINOv2 representation. An independent CLIP image–text baseline tests semantic zero-shot labeling.

Principal finding. Mean pooling achieved 72.78% accuracy / 72.64% macro-F1, versus 69.44% / 69.14% for the single-frame MLP. The GRU reached 71.67% / 71.08%; it did not beat mean pooling. Ten shuffled frame orders averaged 71.78% accuracy. Thus, access to more visual observations helped in this setting, but a benefit from chronological relationships was not demonstrated.

2 Data, integrity and representation

Dataset and evaluation	Verified configuration
HMDB51 subset	brush_hair, clap, drink, eat, pour, wave
Video partitions	600 selected: 420 official train / 180 official test; 70 / 30 per class
Validation	Stratified 336 train / 84 validation, split only from official training videos
Visual observations	8 uniformly sampled RGB frames per video; OpenCV sequential decoding
Pretrained encoder	Frozen facebook/dinov2-small; 384-dimensional final-layer CLS token
Stored tensor	600 × 8 × 384 float32 embeddings; per-video NPZ checkpoints

Extraction and comparison protocol

I first count decodable video frames, then sample eight uniformly spaced indices and decode them sequentially. Frames are converted BGR→RGB and processed by the pretrained encoder in inference mode (batches of four). The resulting 8 × 384 embedding tensor preserves frame order. I use the original dataset split for test evaluation and fit each supervised model's scaler on its training partition only; the model-specific scaler is reused at test time.

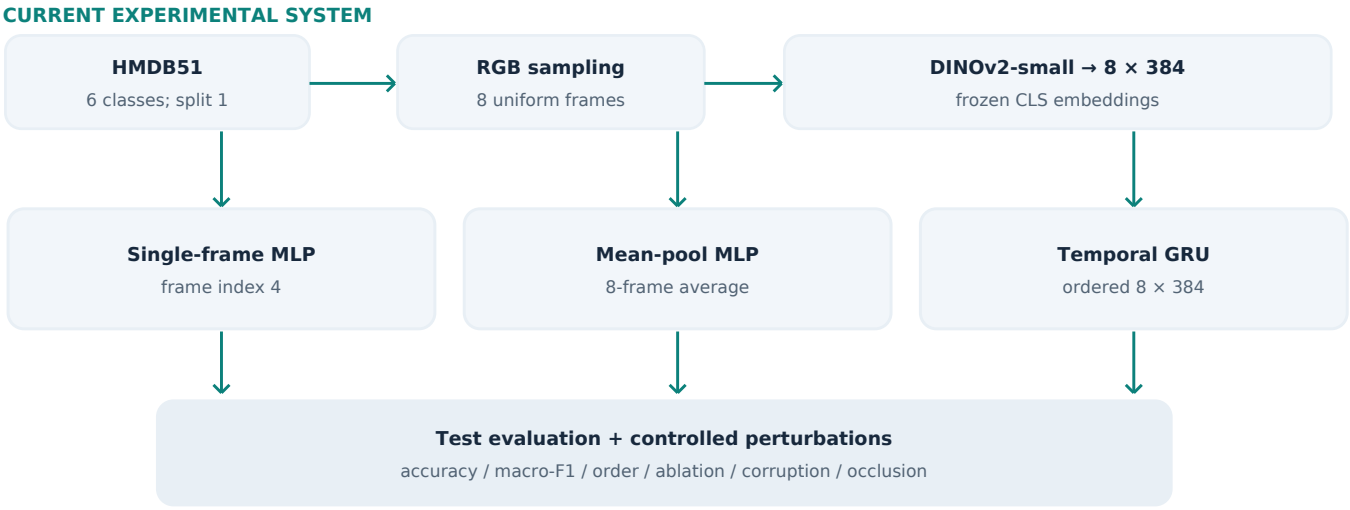


Same official test split (180 videos); frozen visual encoder for supervised comparisons

Research controls: the single-frame MLP sees the middle frame; the mean-pooling MLP sees the average of all eight frame vectors; the GRU sees their ordered sequence. All three use the same DINOv2 feature extraction, isolating a meaningful (though not perfectly architecture-matched) comparison.

3 System architecture and experimental design

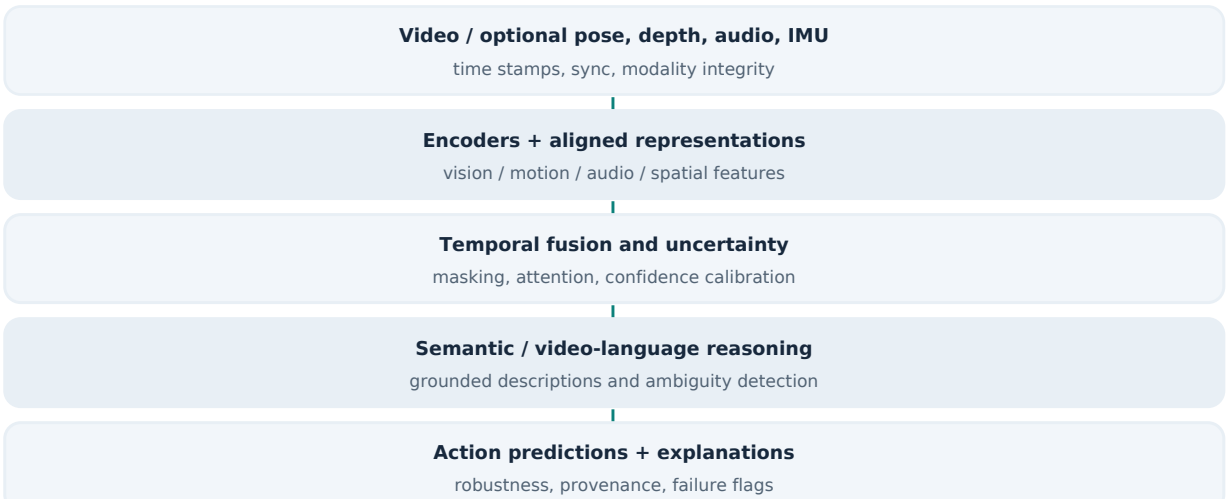
The current pipeline extracts a reusable fixed-length visual representation and evaluates three supervised heads plus a separate CLIP zero-shot pathway. This design separates visual features from the temporal modeling decision.



Model and evaluation decisions

Component	Decision and rationale
Single-frame MLP	One 384D middle-frame vector; reference for image-only evidence
Mean-pooling MLP	Mean of 8 × 384D; eight observations without order dependence
Temporal GRU	One-layer GRU (hidden size 128), followed by classification head; ordered 8 × 384D
CLIP image–text	openai/clip-vit-base-patch32; six action prompts, frame-wise similarities averaged
Evaluation	Official 180-video test: accuracy, macro-F1, class confusion, perturbation sensitivity

Evolution into a multimodal perception system



Proposed extension, not a claim of implementation: synchronize timestamped observations; learn motion-aware features; fuse optional pose/depth/audio/IMU with missing-modality masks; use a temporal attention/fusion encoder; apply a video-language layer for grounded semantic interpretation, ambiguity flags and uncertainty-aware predictions. The current CLIP experiment provides a semantic label baseline, not sequence-level language reasoning.

4 Main model comparison and failure analysis

OFFICIAL TEST ACCURACY (N=180)



Test method	Evidence received	Accuracy	Macro-F1
Single-frame MLP	1 × 384 (middle frame)	69.44%	69.14%
Mean-pool MLP	Mean of 8 vectors	72.78%	72.64%
Temporal GRU	Ordered 8 × 384	71.67%	71.08%
CLIP zero-shot	Mean of 8 similarity scores	71.67%	70.99%

Mean pooling improves accuracy over the single-frame baseline by 3.33 percentage points (macro-F1 +3.50 pp). The GRU trails mean pooling by 1.11 pp accuracy (macro-F1 −1.56 pp). CLIP matches the GRU's accuracy without task-specific training; unlike a sequential VLM, it discards temporal order when averaging scores.

Class-wise recall: correctly classified test clips out of 30

True class	Single frame	Mean pool	GRU
brush_hair	25	27	25
clap	12	13	12
drink	18	20	21
eat	21	22	23
pour	27	27	26
wave	22	22	22
TOTAL / 180	125	131	129

The GRU gains one correct example each for drink and eat relative to mean pooling, but loses on brush_hair, clap and pour. Common GRU confusions include clap → wave (11/30) and drink → eat (8/30); these actions can share static appearance or hand–object cues. Pour is among the easier classes in this subset.

Shared failure cases, not just total accuracy

All three supervised models correctly classify the same 112 clips and misclassify the same 36; the remaining 32 have mixed outcomes. Against mean pooling, the GRU fixes 7 mistakes but introduces 9 errors on previously correct clips. Those complementary and regressed cases argue against claiming a uniform sequence-model advantage.

5 Controlled temporal experiments

5.1 Does the trained GRU use chronological order?

GRU input condition	Accuracy	Macro-F1
Original chronological order	71.67%	71.08%
Reversed order	70.56%	70.17%
10 shuffled orders (mean)	71.78%	71.27%
Shuffles (standard deviation)	1.52 pp	1.47 pp

I evaluated the same trained GRU on one reversed order and ten fixed, randomly permuted eight-frame sequences. Reversal reduces accuracy by 1.11 pp; shuffled orders range from 70.00% to 75.00% and average marginally above original order. The evidence does not establish consistent reliance on the true chronology. These are input perturbations, not retraining runs, and their standard deviation is not an uncertainty interval for model training.

5.2 Progressive observation: two distinct experiments

Experiment A — first K chronological frames. Separately trained GRUs receive prefixes of length 1, 2, 4 or 8.

Frames K	1	2	4	8
Accuracy	64.44%	70.00%	70.56%	72.78%
Macro-F1	64.04%	69.72%	70.05%	72.27%

Experiment B — coverage-controlled selection. Separate GRUs use strategically spaced indices to distribute the observations through the clip.

Frames	Indices	Accuracy	Macro-F1
1	[4]	73.89%	73.64%
2	[0,7]	70.56%	69.74%
4	[0,2,5,7]	73.89%	72.98%
8	[0,1,2,3,4,5,6,7]	72.22%	71.26%

The first-K curve improves as the observation window grows, whereas coverage-controlled accuracy is non-monotonic. Both studies compare separately trained models, not an online rollout of the same model. The progressive GRUs have a somewhat different classification head from the primary GRU; therefore, their 8-frame results must not be treated as repetitions of the main 71.67% result, and their one-frame score is not a controlled MLP comparison.

5.3 Frame-level embedding ablation

At inference, I individually replace one standardized GRU frame vector by zero (the training-set mean feature vector in standardized coordinates). Baseline accuracy is 71.67%. Removing frames 5 or 6 gives 73.89% (+2.22 pp each); removing frame 7 gives 73.33% (+1.67 pp). Frame 1 removal gives 71.11%. This is sensitivity to one substituted vector, not proof those frames intrinsically lack useful motion information.

6 Robustness, representation and multimodal probe

6.1 Visual corruption and spatial occlusion

Test-time condition	Accuracy	Macro-F1	Δ Accuracy
Unmodified frames	71.67%	71.08%	—
Gaussian blur (radius 2)	69.44%	69.07%	−2.22 pp
Gaussian noise (σ=20/255)	67.78%	67.40%	−3.89 pp
Brightness × 0.5	74.44%	74.18%	+2.78 pp
Central black rectangle (25% W × 25% H)	68.33%	67.54%	−3.33 pp

The same trained DINOv2–GRU and test set are used. Corruption and occlusion are applied to sampled RGB images before embedding extraction. Gaussian noise creates the largest observed drop; blur and central occlusion also harm predictions. Halving brightness unexpectedly helps on this sample and severity, which may reflect an incidental feature shift rather than general robustness. Only one severity/location was tested per intervention.

6.2 Temporal embedding reasoning

Each video is represented as an ordered trajectory E ∈ R^(8×384). This representation allows adjacent-frame cosine similarity, inter-video similarity, and frame-wise variation to be examined independently of the classifier. The repository's temporal embedding-analysis pathway is designed for those diagnostics. This report does not assign numerical cosine-similarity trends or claim specific trajectories without their underlying analysis artifacts. The classification experiments alone cannot establish which representation changes correspond to true motion.

6.3 Vision-language probe: what value does CLIP add?

For each of 180 test videos, I compare eight CLIP frame embeddings against six text prompts such as "A video of a person pouring a liquid." I average class similarity scores across frames and take the highest-scoring class; all 180 videos were evaluated, with zero recorded failures. Accuracy is 71.67% and macro-F1 70.99%. This provides interpretable natural-language category definitions and a competitive zero-shot semantic reference. It does not produce sequence descriptions, explain transitions or distinguish temporal direction; stronger video-language models would be required to test those capabilities.

6.4 What can and cannot be inferred

Because the encoder is frozen, gains from multi-frame pooling reflect how pretrained appearance features are aggregated rather than end-to-end motion feature learning. Accuracy is measured on only 180 test clips from six selected classes and one official split. A difference of one correct prediction equals about 0.56 percentage points; the small gaps here should not be overstated without repeated seeds, additional splits, and paired statistical analysis.

7 Direct answers to the assessment research questions

A — When did temporal information help?

Additional observations helped: eight-frame mean pooling was +3.33 pp more accurate than a single representative frame. Class-wise, the GRU gained one correctly recognized drink clip and one eat clip versus mean pooling. The first-K progression also improved from 64.44% (one frame) to 72.78% (eight), though those are separately trained GRUs. None of these facts alone proves use of chronological motion patterns.

B — When did it not help?

Explicit recurrence did not beat unordered pooling overall (71.67% versus 72.78%). Shuffling the eight frames averaged 71.78%, and in the coverage-controlled experiment one- and four-frame settings matched at 73.89%, exceeding the eight-frame variant. Some categories, especially clap and wave, retained substantial confusion.

C — How to isolate true temporal relationships from more images?

Hold frame identities, embeddings, split, parameter budget and training protocol fixed; compare order-sensitive modeling against order-invariant pooling, then disrupt order using reversals and multiple shuffles without changing visual content. Test on action pairs that are distinguishable primarily by direction or transitions, and repeat across seeds. Here, both the mean-pool result and weak order perturbation response caution against attributing gains to chronological reasoning.

D — What would falsify temporal usefulness for this selected task?

Repeated trials where a properly tuned, comparable temporal model fails to outperform order-invariant aggregation on held-out clips; predictions remain nearly unchanged under systematic order disruption; and motion-dependent subset performance shows no gain. The current evidence trends in that direction but is not sufficiently replicated to be decisive.

E — What would strengthen the evidence?

Add temporally diagnostic action pairs, more categories and HMDB51 splits; use motion-aware or denser frame sampling, pose/optical-flow cues and matched-capacity temporal baselines. Repeated seeded runs, confidence intervals/paired tests, explicit ambiguity annotations, and frame-strip examples would help separate object appearance, observation coverage and motion understanding.

8 Limitations, near-term priorities and conclusion

Limitations. Six classes and 600 videos; 8 uniform frames may miss decisive motion; one principal GRU architecture and one official test split; small, unmatched differences in model heads/training; limited corruption severity/occlusion location; only ten shuffle permutations; single-frame MLP training not fully seeded. CLIP score averaging is semantic but not a temporal video-language evaluation.

Next two days, prioritized. (1) Add temporally diagnostic sequences and motion-aware frame selection with matched pooling/GRU controls. (2) Repeat across seeds, test splits and paired significance analyses. (3) Improve experiment configuration, unit tests, reproducible frame-strip/embedding plots and error-case documentation.

Conclusion. This experiment supports a modest benefit from multiple views of a video, not a verified advantage from modeling their chronological structure. The result is scientifically useful: better observation coverage is not the same thing as learned temporal reasoning.

Research artifacts: supervised metrics and confusion matrices; temporal-order and progressive-observation JSON; CLIP prompts/results; frame ablation, corruption, occlusion and error-comparison outputs. See the accompanying architecture.md and repository README for implementation pathways.